

GEORGIA SOUTHERN UNIVERSITY
ALLEN E. PAULSON COLLEGE OF ENGINEERING AND COMPUTING
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

An Integrated Biomimetic Framework for High-Efficiency AUVs

Anish Goyal
901414697

Dr. Min Gyu Kim
Assistant Professor

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Abstract

This proposal serves as an introduction to my research in biomimetic underwater robotics in preparation for the GRFP application. The proposal outlines a multi-year graduate research plan to develop and validate an integrated biomimetic framework for high-efficiency autonomous underwater vehicles; although, the GRFP does not mandate that I actually carry out this research while in graduate school. Nevertheless, I intend to continue this research before I graduate.

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1 Introduction

Autonomous underwater vehicles (AUVs) enable important missions in oceanography and climate research [1], [2]. Current AUVs use conventional torpedo-shaped hulls and propellers. This design limits their energy efficiency and operational range [2].

Bio-inspired robotics offers a path to innovation. Researchers have built AUVs that mimic fast swimmers like tuna and manta rays [3], [4], [5]. These designs improve speed and maneuverability. A research gap still exists. Most designs copy animal morphology, or shape. They neglect the neuromuscular control that makes these animals so efficient [5], [6]. This creates AUVs that look like fish but lack their energy-saving adaptations. The challenge is to build an integrated system. It must combine a hydrodynamically optimal hull with a neurologically-derived propulsion controller.

I propose a research plan to develop and test such a system. My framework uses a dual-optimization strategy. First, a topological model generates a hull shape from the Navier-Stokes equations to minimize drag. Second, a neurological control model uses the synaptic firing patterns of Zebrafish (*Danio rerio*), shown in Figure 1, to create energy-efficient motor commands. This integrated approach will improve AUV endurance and adaptability. It will enable long-duration missions to monitor deep-sea ecosystems affected by climate change [7].



Figure 1: Zebrafish photo by [Massachusetts Institute of Technology Brain and Cognitive Sciences](#)

2 Intellectual Merit

This research builds on a foundational project my team and I completed. That work validated the core concepts through experiments and statistical analysis [8]. My proposed work contains three key innovations.

2.1 Computationally-Derived Topology for Hydrodynamic Supremacy

Traditional AUV design is slow and may not find the best solution. I will generate the hull topology from a computational fluid dynamics (CFD) simulation. This optimizes the design for its environment from first principles [9].

My method uses the Navier-Stokes equations, shown in Equation 1 and Equation 2, to model fluid motion [10].

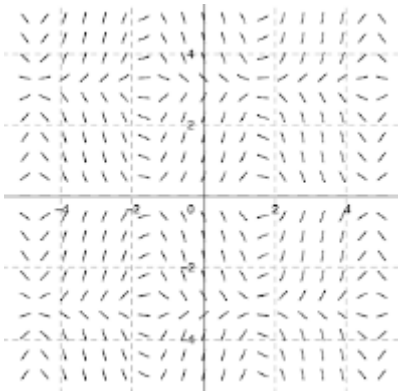
$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f} \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0 \quad (2)$$

Here, $\mathbf{u}$ is the velocity field, ρ is fluid density, p is pressure, and ν is kinematic viscosity. I will numerically solve these equations to create an optimized 3D velocity field. This field informs a hull geometry that minimizes total hydrodynamic drag force, F_{drag} , calculated with Equation 3.

$$F_{drag} = \int_{\partial\Omega} (-p\mathbf{n} + \nu \frac{\partial \mathbf{u}}{\partial n}) \cdot \mathbf{n} dA \quad (3)$$

In our foundational work, we optimized the model generation process. Standard 3D grid graphing was too slow. We implemented a heap-based data structure and an octree indexing algorithm. This approach partitions 3D space into eight quadrants. It reduces search operations and lowers computational overhead for model generation.



(a) CFD vector field



(b) 3D-printed prototype

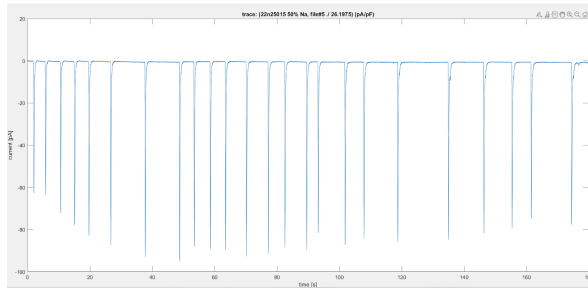
Figure 2: The topological model is derived from a CFD simulation (Figure 2a) and realized as a 3D-printed prototype (Figure 2b).

2.2 Functional Bio-integration via Neuromuscular Control

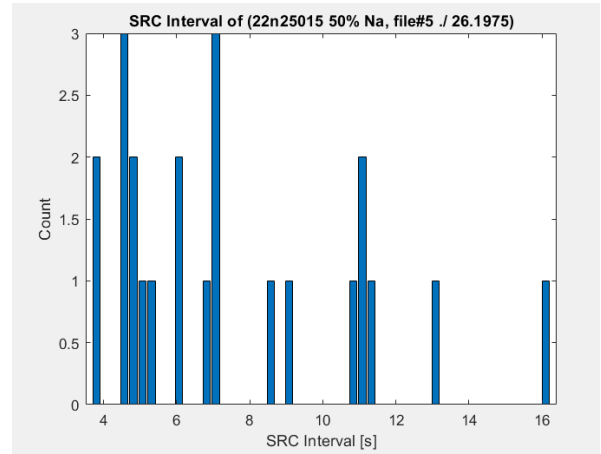
My research achieves functional bio-integration. It adopts the control logic of a biological system. The propulsion system uses synaptic data from developing Zebrafish from the Tuan Bui laboratory [11]. Zebrafish are an ideal model. They have well-understood spinal cord circuitry and use Central Pattern Generators (CPGs) for rhythmic swimming. This allows the study of motor control without sensory inputs.

In our foundational work, we developed a MATLAB pipeline to process raw electrocardiogram (ECG) signals from this dataset. The process involved three steps:

1. **Normalization:** We normalized raw signal traces to remove baseline noise.
2. **Peak Identification:** We used the `findpeaks` function to identify significant synaptic events corresponding to muscle activation.
3. **Control Signal Abstraction:** We created a metric called the “Synaptic Value” from the frequency of validated peaks. This converts a noisy biological signal into a clean command for a robotic control system.



(a) Normalized ECG data



(b) Identified peaks in ECG data

Figure 3: Raw ECG data from the Bui Lab (Figure 3a) is processed using a `findpeaks` algorithm to identify locomotor events (Figure 3b).

2.3 Synergistic Integration in a Closed-Loop System

The novelty of this framework is the coupling of the optimized body with its neurological model. We achieved this integration using the Robot Operating System (ROS). A ROS node subscribed to the “Synaptic Value” from our MATLAB script. It then published speed commands to the motor driver. This created a closed-loop system where a biological signal continuously modulated propulsion.

This architecture acts as a bio-inspired Variable Frequency Drive (VFD). VFDs are a key part of industrial energy efficiency [12], [13], [14]. They match motor speed to real-time demand. Our neurological controller does the same function. It uses an evolutionarily optimized control signal for efficient aquatic locomotion.

Our preliminary research validated this approach. Controlled experiments, summarized in Table 1, showed significant improvements in traversal time and energy efficiency for the integrated system. These results provide a solid foundation for my proposed graduate research.

Table 1: Summary of Foundational Experimental Results [8]

Performance Metric	Group	Mean	Std. Dev.	Z-Score	P-Value
Traversal Time (s) (Topological Model)	Control	3.548	0.031	Ref.	-
	Modified Design	3.464	0.024	-2.71	0.003
Energy Consumption (kWh) (Propulsion Model)	Control	0.0590	0.0020	Ref.	-
	Modified Design	0.0530	0.0016	-3.09	0.001

3 Research Objectives

I have four primary objectives for my graduate research. Each objective builds upon our preliminary work. They advance the system toward a fully autonomous, adaptive AUV.

3.1 Data Collection with Live Vertebrate Animals

The existing Tuan Bui lab dataset [11] is valuable but limited. It lacks synchronized data on ambient water conditions such as temperature, salinity, and viscosity. To build a truly adaptive neurological controller, I need a richer dataset. This objective focuses on collecting new ECG data from live zebrafish under varied and controlled environmental conditions.

3.1.1 Methodology

1. **Experimental Setup:** I will establish a dedicated aquatic laboratory environment. This setup will include a housing system for zebrafish and a testing tank equipped with sensors and actuators. These will control and monitor water temperature, salinity, viscosity, and flow patterns.
2. **Data Acquisition:** I will record ECG signals from zebrafish while systematically altering the water parameters. High-speed cameras will simultaneously track the fish's locomotion. This will create a comprehensive dataset that links neural signals, movement, and specific environmental conditions.
3. **Protocol and Compliance:** All experiments will be designed to minimize animal stress. I will submit a detailed research protocol to the Institutional Animal Care and Use Committee for review and approval before any experimentation begins.

3.2 Advance the Neurological Controller from a Heuristic to a Predictive System

The current findpeaks algorithm is reactive. It responds to a synaptic spike after it occurs. I will transform this into a predictive system that anticipates propulsion needs.

Recurrent Neural Networks (RNNs) are good for time-series data. Long Short-Term Memory (LSTM) networks are a specialized RNN type. They are powerful for modeling complex biological signals like ECGs [15], [16], [17]. I will replace the reactive algorithm with a predictive LSTM model.

3.2.1 Methodology

1. **Data Preparation:** I will use the new dataset to structure overlapping sequences for supervised time-series forecasting [18], [19].
2. **Model Development:** I will implement an LSTM-based RNN in Python using TensorFlow/Keras. The network architecture is shown in Figure 4. I will compile the model using an Adam optimizer and a mean squared error loss function [20], [21].
3. **Validation & Integration:** I will benchmark the LSTM model's accuracy against the original method. I will then deploy the trained model within the ROS framework to provide predictive motor control.

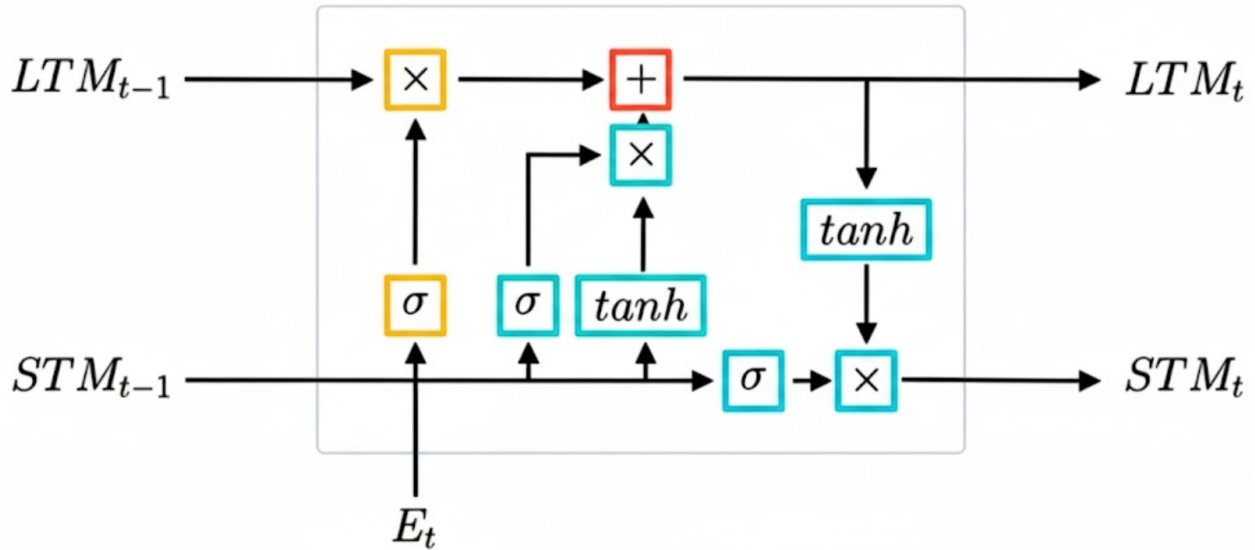


Figure 4: LSTM network architecture for predictive neurological control

3.3 Refine the Topological Model for Dynamic and Complex Marine Environments

The current topological model is static. It is optimized for a single set of fluid parameters. Real marine environments are dynamic. I will transform the static model into a dynamic physics engine to inform the AUV's control strategy in real-time.

3.3.1 Methodology

1. **Model Enhancement:** I will extend the C++ CFD model. It will accept real-time inputs for water density and viscosity. I will also incorporate turbulence models to simulate realistic ocean conditions.
2. **Simulation and Validation:** I will conduct simulations in MATLAB Simulink. These simulations will model the AUV's performance across a range of environmental conditions. This will generate a performance map shown in Figure 5.
3. **Physical Prototype V2.0:** The simulation results will inform a second-generation AUV hull. I will use advanced 3D printing technologies like Selective Laser Sintering (SLS) for fabrication.

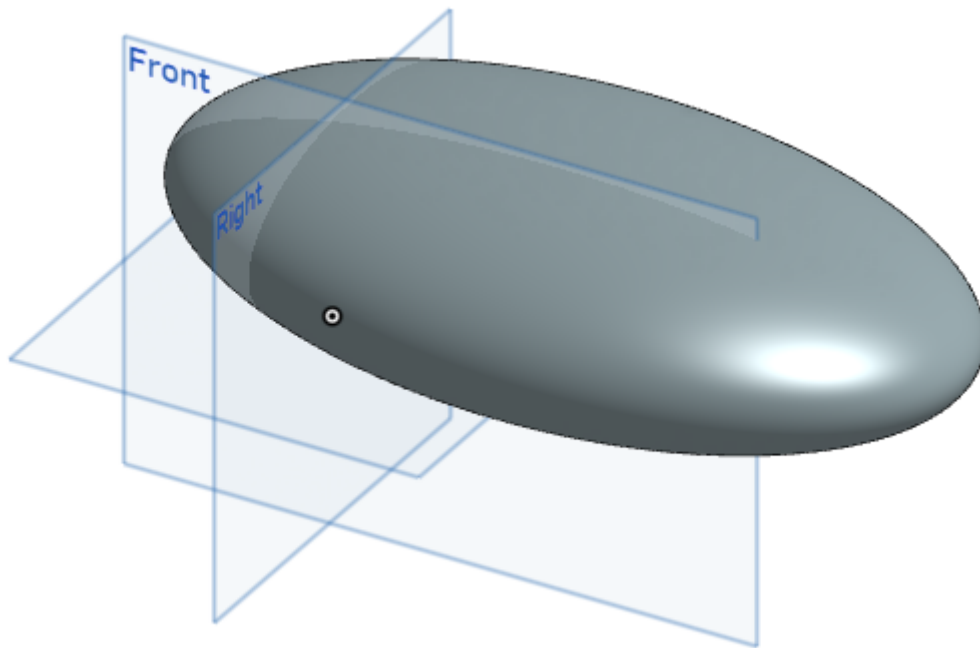


Figure 5: MATLAB Simulink model for AUV hydrodynamic performance

3.4 Integrate Multi-Sensor Feedback for Full-Loop Autonomous Operation

My final objective is to achieve true autonomous operation. This requires closing the loop between environmental sensing, dynamic modeling, and predictive control. I will instrument the AUV with sensors and expand the ROS framework.

3.4.1 Methodology

1. **Sensor Suite Integration:** I will instrument Prototype V2.0 with environmental sensors. These include salinity, temperature, and pressure sensors.
2. **LIDAR for Environmental Mapping:** I will integrate a compact LIDAR module into the AUV. This sensor will provide a real-time 3D point-cloud of the AUV's surroundings for obstacle avoidance and mapping.
3. **ROS Framework Expansion:** I will expand the ROS architecture to include nodes for each new sensor. A central “mission controller” node will fuse the data. This controller will use the performance map to dynamically adapt the LSTM controller's parameters. This final integration is shown in Figure 6.

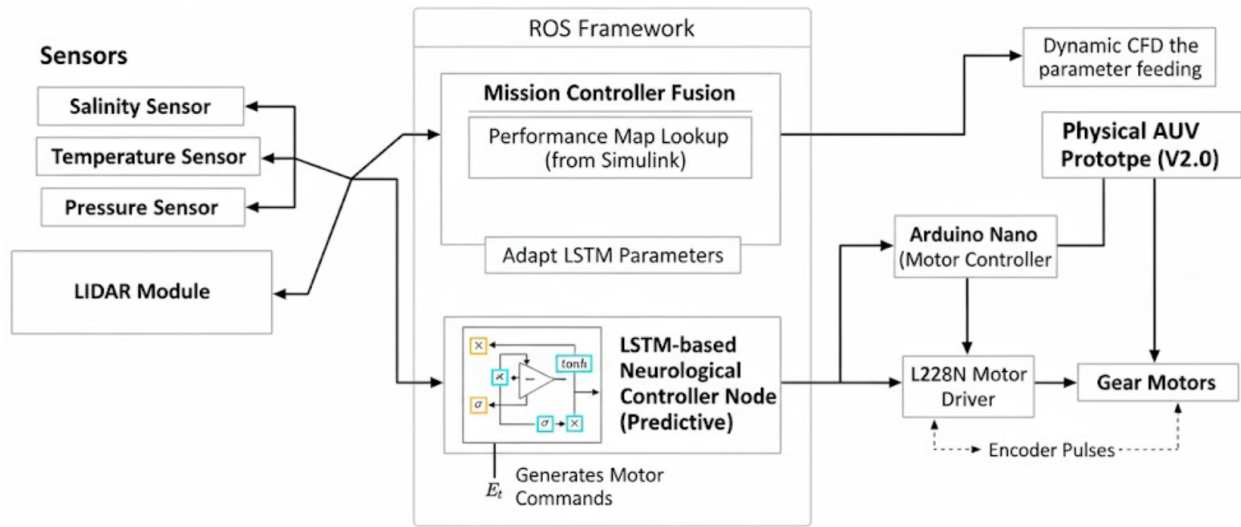


Figure 6: Full system architecture for AUV autonomy

4 Broader Impacts

This research will yield significant impacts. It addresses key scientific, societal, and educational priorities.

- **Contribution to Science and Engineering:** This work will advance biomimetic robotics, computational fluid dynamics, and applied machine learning. I will make the developed framework available as an open-source platform.
- **Societal Benefits:** This research supports critical national and global objectives.
 - **Climate Change Research:** Long-duration missions will enable persistent monitoring of sensitive deep-sea ecosystems [7].
 - **Ocean Exploration and Economy:** The technology will lower the cost of underwater mapping.
 - **Infrastructure and Security:** The AUV will serve as a platform for inspecting critical underwater infrastructure.
- **Educational Outreach:** The project is an ideal vehicle for STEM outreach. I will develop an educational module to engage high school students.

5 Budgetary Requirements and Resource Feasibility

This project is highly feasible within a university setting. Core computational work can use existing campus computing clusters. The initial prototype was built from low-cost components. However, the new data collection phase requires specialized equipment and may require travel to a collaborating institution with appropriate facilities.

I am requesting funding to support the live animal experiments, acquire advanced sensing components, and disseminate research findings. A detailed cost breakdown is in Table 2.

Table 2: Proposed Budget for Graduate Research

Category	Item	Quantity	Unit Cost (USD)	Total Cost (USD)	Justification
Zebrafish Experimentation	Zebrafish Housing System	1	\$2,000	\$2,000	Recirculating aquatic housing for live animals [1, 2]
	Wild-Type Zebrafish	20 pairs	\$25	\$500	Initial colony for breeding and experimentation [3]
	Digital Viscometer	1	\$750	\$750	Measurement of water viscosity for experiments [4, 5]
Materials & Supplies	100-Gallon Water Tank	1	\$350	\$350	Larger testing environment for advanced maneuvers [22], [23]
	Research-Grade Salinity Sensor	1	\$250	\$250	Real-time data for dynamic CFD model [24], [25]
	Digital Temperature Probe	1	\$80	\$80	Real-time data for dynamic CFD model [26]
	Pressure Sensor	1	\$50	\$50	Accurate depth measurement
	Low-Cost 3D LIDAR Module	1	\$200	\$200	Obstacle avoidance and mapping

Category	Item	Quantity	Unit Cost (USD)	Total Cost (USD)	Justification
Travel	Advanced 3D Printing Materials (SLS/TPU)	-	\$500	\$500	Fabrication of Prototype V2.0
	Electronic Components (Microcontrollers, etc.)	-	\$300	\$300	Upgrades and replacements
	External Laboratory Travel	1 trip	\$1,500	\$1,500	For experiments requiring specialized facilities [6, 7] Dissemination of results to a specialized audience
	Conference Travel	2	\$1,500	\$3,000	
Publication Fees	Open-Access Journal Fees	2	\$2,000	\$4,000	Ensuring broad accessibility of research findings
Total				\$15,480	

5.1 Statement on Ethical Requirements for Vertebrate Animals

I have prepared this proposal with careful consideration of the National Science Foundation's guidelines on research involving vertebrate animals [27], [28].

This research will involve live vertebrate animals. The initial neurological model development used a pre-existing, publicly available dataset from the Tuan Bui laboratory [11]. However, this dataset lacks the synchronized environmental data needed to train a truly adaptive control system. To overcome this limitation, I will conduct new experiments to collect ECG and locomotion data from zebrafish under a variety of controlled aquatic conditions.

All animal procedures will be designed to minimize pain and distress. I will submit a detailed research protocol to the Institutional Animal Care and Use Committee (IACUC) for review and approval before beginning any experiments. This ensures my research adheres to the "3Rs" principles (Reduce, Refine, Replace) and complies with all federal and institutional guidelines for ethical research conduct [29], [30], [31], [32].

References

- [1] “Remote sensing | special issue : Advances on autonomous underwater vehicles (AUV).” [https://www.mdpi.com/journal/remotesensing/special_issues/autonomous_underwater_vehicles%5D\(https://www.mdpi.com/journal/remotesensing/special_issues/autonomous_underwater_vehicles\)](https://www.mdpi.com/journal/remotesensing/special_issues/autonomous_underwater_vehicles%5D(https://www.mdpi.com/journal/remotesensing/special_issues/autonomous_underwater_vehicles)).
- [2] “Autonomous underwater vehicle - wikipedia.” [https://en.wikipedia.org/wiki/Autonomous_underwater_vehicle%5D\(https://en.wikipedia.org/wiki/Autonomous_underwater_vehicle\)](https://en.wikipedia.org/wiki/Autonomous_underwater_vehicle%5D(https://en.wikipedia.org/wiki/Autonomous_underwater_vehicle)).
- [3] I. Mitin, R. Korotaev, A. Ermolaev, V. Mironov, S. A. Lobov, and V. B. Kazantsev, “Bioinspired propulsion system for a thunniform robotic fish,” *MDPI*, 2022, doi: [10.3390/biomimetics7040215](https://doi.org/10.3390/biomimetics7040215).
- [4] P. Krishnamurthy, F. Khorrami, J. de Leeuw, M. E. Porter, K. Livingston, and J. H. Long Jr., “An electric ray inspired biomimetic autonomous underwater vehicle,” in *2007 IEEE/RSJ international conference on intelligent robots and systems*, 2007, pp. 1336–1341. doi: [10.1109/IROS.2007.4399201](https://doi.org/10.1109/IROS.2007.4399201).
- [5] H. Li, S. Wang, and J. Yang, “Underwater undulating propulsion biomimetic robots: A review,” *Biomimetics*, vol. 8, no. 3, p. 318, 2023, doi: [10.3390/biomimetics8030318](https://doi.org/10.3390/biomimetics8030318).
- [6] R. Uppal, “Innovation beneath the waves: Biomimetic propulsion systems for ships and unmanned underwater vehicle.” *International Defense Security & Technology*, May 2024.
- [7] K. Beem, “Rolling in the deep: Climate change and deep sea ecosystems.” *Columbia University*, 2020.
- [8] A. Goyal, D. Iliev, and R. Guardado, “A novel underwater biomimetic propulsion system derived from zebrafish neurological analysis,” in *2025 IEEE underwater technology (UT)*, 2025, pp. 1–5. doi: [10.1109/UT61067.2025.10947443](https://doi.org/10.1109/UT61067.2025.10947443).
- [9] Z. Zhang, Q. Wang, and S. Zhang, “Review of computational fluid dynamics analysis in biomimetic applications for underwater vehicles,” *Biomimetics*, vol. 9, no. 2, p. 79, 2024, doi: [10.3390/biomimetics9020079](https://doi.org/10.3390/biomimetics9020079).
- [10] A. Evgrafov, “Topology optimization of navier-stokes equations,” *Technical University of Denmark*, 2004.
- [11] Y. Roussel, S. F. Gaudreau, E. R. Kacer, M. Sengupta, and T. V. Bui, “Modeling spinal locomotor circuits for movements in developing zebrafish,” *eLife*, vol. 10, p. e67453, 2021, doi: [10.7554/eLife.67453](https://doi.org/10.7554/eLife.67453).
- [12] “Understanding energy-efficient submersible pump for cost-saving solutions.” *Sleka Ind.*
- [13] “Energy efficiency tips for submersible pumps.” *Btali International*.
- [14] “Modern submersible pumps: How they reduce energy costs.” *Unnati Pumps*.
- [15] B.-H. Kim and J.-Y. Pyun, “ECG identification for personal authentication using LSTM-based deep recurrent neural networks,” *Sensors*, vol. 20, no. 11, p. 3069, 2020, doi: [10.3390/s20113069](https://doi.org/10.3390/s20113069).
- [16] V. Felizardo and N. M. Garcia, “ECG forecasting system based on long short-term memory,” *Bioengineering*, vol. 11, no. 1, p. 89, 2024, doi: [10.3390/bioengineering11010089](https://doi.org/10.3390/bioengineering11010089).
- [17] S. K. Yadav, A. Tiwari, A. Singh, and J. Kumar, “Deep conv-LSTM network for arrhythmia detection using ECG data,” *International Journal of Advanced Computer Science and Applications*, vol. 14, no. 9, 2023, doi: [10.14569/IJACSA.2023.0140973](https://doi.org/10.14569/IJACSA.2023.0140973).
- [18] “Recurrent neural network tutorial (RNN).” *DataCamp*.
- [19] “RNN PyTorch time series tutorial: Complete guide to implementation.” *Codecademy*.

- [20] “Classify ECG signals using long short-term memory networks.” MATLAB & Simulink.
- [21] P. Venkatesan, A. Bhardwaj, and J. Sivaraman, “Design and development of LSTM-RNN model for the prediction of RR intervals in ECG signals,” in *International conference on innovative computing and communications*, 2021, pp. 133–141.
- [22] “100 gallon plastic water & liquid storage tanks.” Plastic-Mart.
- [23] “100 gallon fresh / gray water holding tank.” Class A Customs.
- [24] “Salinity sensor.” Vernier.
- [25] “Salinity sensors: Top suppliers & models for accurate water testing.” Accio.
- [26] “Digital thermometers for laboratory use: calibratable.” Cole-Parmer.
- [27] “Research involving vertebrate animals.” National Science Foundation.
- [28] “Proposal & award policies & procedures guide (PAPPG).” National Science Foundation.
- [29] “Institutional animal care & use committee.” University of Michigan.
- [30] “Excerpted from: The national science foundation grants policy manual.” Millersville University.
- [31] “Office of the institutional animal care and use committee.” University of Iowa.
- [32] “The IACUC.” Office of Laboratory Animal Welfare (OLAW).
- [33] A. A. Saputra, J. Botzheim, and N. Kubota, “Neuro-cognitive locomotion with dynamic attention on topological structure,” *MDPI*, 2023, doi: [10.3390/machines11060619](https://doi.org/10.3390/machines11060619).
- [34] A. Thomas, K. Bates, A. Elchesen, I. Hartsock, H. Lu, and P. Bubenik, “Topological data analysis of c. *Elegans* locomotion and behavior,” *Frontiers in Applied Mathematics and Statistics*, 2021, doi: [10.3389/frai.2021.658269](https://doi.org/10.3389/frai.2021.658269).